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Iron and niobium based additives in magnesium hydride: Microstructure and hydrogen storage properties

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ABSTRACT

In this study, powder mixtures of MgH_2 + 2 mol.% X, with X = Nb, Nb_2O_5, NbF_5, Fe, Fe_2O_3, FeF_3, were processed by mechanical milling at liquid nitrogen temperature (cryomilling). The effect of additives on crystalline structure, thermal properties and hydrogen storage properties of the mixtures were investigated. Morphological investigations indicated a heterogeneous particle size distribution of the powder mixtures and a fine dispersion of additive particles (FeF_3) in the MgH_2 matrix. High resolution synchrotron radiation X-ray diffraction (SR-XRD) data followed by Rietveld refinements showed a significant reduction on crystallite size for the samples containing fluorides (11 nm) in comparison with the pure MgH_2 sample (29 nm). This was related to the mechanical behavior of fluorides during milling with MgH_2, which act as a lubricant, dispersing and/or cracking agent during milling, and thus helping to further reduce MgH_2 particle size. DSC analysis revealed that fluorides (NbF_5, FeF_3) are much more effective than oxides (Nb_2O_5, Fe_2O_3) and the transition metals (Nb and Fe), respectively, in reduction the desorption temperature. Furthermore, Nb_2O_5 is more efficient than Fe_2O_3. Finally, the best results for desorption kinetics were observed for the fluorides: NbF_5 and FeF_3 (equivalent effect and consistent to the DSC analysis) followed by the oxides: Nb_2O_5, Fe_2O_3 and Nb. The addition of Fe was not efficient in comparison with the pure cryomilled sample.

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Introduction

MgH_2 is a promising candidate for hydrogen storage applications due to the high reversible hydrogen gravimetric capacity (7.6 wt. %), high volumetric density (110 g/l) as well as low cost of Mg [1]. However, its main limitations for practical applications are the slow hydrogen absorption/desorption kinetics and the high thermodynamic stability [2,3].

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The hydrogen storage properties of MgH_2 can be remarkably improved by high-energy ball milling (HEBM) [4–9], which can be performed under argon or hydrogen atmosphere (reactive milling). In addition to the increase in the specific surface area, raised concentration of defects and significant reduction on the crystallite size, HEBM is also suitable to introduce a fine dispersion of additives (so-called "*catalysts*") with catalytic effect.

In the last decades, a several numbers of studies have successfully demonstrated the positive effects of additives and their strong effect on the hydrogen sorption kinetics of MgH_2. The most common additives are based on transition metals (e.g.: Fe, Nb, Ni, Ti, V, Ce, Co etc.), their oxides (e.g.: Nb_2O_5; Fe_2O_3) and fluorides (e.g.: FeF_3, TiF_3, NiF_2, NbF_5) [5–14]. Added these elements/compounds at low proportions into MgH_2 system during the HEBM process favor the chemical reaction with hydrogen through the creation of nucleation sites by the presence of new interfaces, in addition to decreasing the metal-hydrogen bonds energy and reducing the desorption energies of MgH_2 [15]. By using additives in MgH_2, very fast hydrogen absorption/desorption kinetics behavior (around minutes or seconds) and a significant reduction on desorption temperature (around 250 °C) are consistently reported [10,16].

Many studies have also reported the positive effects of iron additions into MgH_2. Since Fe and Mg are immiscible, iron additions should promote the hydrogen desorption via extended defects and creation of extra interfaces onto MgH_2 [8,17]. In the case of niobium-based additives, the improvement into hydrogen storage properties of MgH_2 system after HEBM experiments has been understood by the formation of intermediate phases (NbH_X), which could acts as a gateway for the hydrogen, favoring the desorption kinetics of MgH_2 [9–11]. The presence of nano-interfaces between the phases NbH_x/MgH_2 play a significant role on the kinetics process because these regions possess high diffusivity pathways for the hydrogen. The presence of NbH_x phase was also been observed when MgH_2 was ball milled with Nb additives for 30 min under argon atmosphere [18]. Among the niobium based additives, the Nb_2O_5 has been extensively studied and it is claimed to give the best kinetics properties of MgH_2 [10,16]. Furthermore, it was also reported the beneficial effect of Nb_2O_5 into Mg by the formation of Mg–Nb oxide [19].

Several works have reported substantial kinetic improvements that occur when transition metal fluorides (especially FeF_3 and NbF_5) are used as additives into MgH_2 [12–14]. During milling of the MgH_2-fluoride mixtures or at latest after the first hydrogen desorption, MgH_2 reacts with the fluoride producing MgF_2, H_2 and transition metal or its hydride (depending on the nature of transition metal). On the other hand, these materials usually present greatly enhanced hydrogen absorption and desorption kinetics, though the specific cause for the catalytic effect has not been completely understood up to now. Part of the catalytic effect has been attributed to the transition metal or its hydride, and part to MgF_2 or the F- ion [14,20].

Despite the positive chemical effects on MgH_2/Mg provided by using additives during high-energy ball milling responsible for the kinetics improvements, some interesting mechanical effects are also been reported. In this respect, the oxides and fluorides additives have gained much more attention, especially due to their hardness nature. These additives act as a lubricant, dispersing and/or cracking agent during milling, helping to further reduce the MgH_2 particle size or via a pinning effect, limiting the grain growth during reactions [21–23]. In contrast, metallic catalysts may be more difficult to disperse due to their ductile behavior limiting the fracture of fine particles [7].

In this study, we examined the catalytic effect of two groups of additives: iron group (Fe, Fe_2O_3 and FeF_3) and niobium group (Nb, Nb_2O_5, NbF_5) added into MgH_2 at the proportion of 2 mol.%. The mixtures were produced by milling at liquid nitrogen temperature (named here as: cryomilling). This processing route was chosen mainly by two reasons: a) the possibility to produce nanostructured composites with very small particle size using shorter milling times; b) to test the effects on kinetics properties and the distribution of selected additives into MgH_2 under milling at liquid nitrogen temperature. It is worth noting that, the mixtures proposed here have not been investigated by using this processing route earlier. In fact, the cryomilling route has so far only been used to prepare the following mixtures: 2 Mg + Fe, $LiNiH_2$–LiH_2, MgH_2 + $BaCa_{1-x}Nd_xO_{3-\eta}$ and Nb_2O_5 with hydrogen storage purposes [24–27]. Here, we report structural characteristics and hydrogen storage properties of the different mixtures produced by cryomilling in comparison with literature data.

Experimental

The commercial powders used in this study are: MgH_2, Fe, Nb, Fe_2O_3, Nb_2O_5, FeF_3 and NbF_5. Details on the manufacturer, the purity and the powder size from the starting materials are presented in Table 1.

MgH_2 was initially mixed with the additives X (X = Fe, Fe_2O_3, FeF_3, Nb, Nb_2O_5, NbF_5) at a proportion of 2 mol.%. The mixtures were then milled in a SPEX 6770 Freezer/Mill using a specially designed stainless steel vial containing a cylindrical impactor, which is immersed in liquid nitrogen. Milling was performed for 3 h under Ar atmosphere at an impact frequency of 30 Hz. The mass ratio between the impactor and the powder was of 16:1. Pure MgH_2 without any additive was also milled in the similar conditions for comparison purposes. All powder mixtures before and after milling were carefully

Table 1 – Manufacture, purity (%) and powder size (μm) of the starting materials.

	MgH_2	Fe	Nb	Fe_2O_3	Nb_2O_5	FeF_3	NbF_5
Manufacturer	ABCR	Alfa Aesar	Alfa Aesar	Alfa Aesar	Aldrich	Aldrich	Aldrich
Purity (%)	98	99,98	99,8	99	98	98	98
Powder size (μm)	0,3	44	1–5	350	>200	>100	–

handled in a glove box in an inert Ar atmosphere in order to minimize powder oxidation.

The milled mixtures were structurally characterized by X-ray diffraction (XRD) using two X-rays sources: (a) for preliminary analysis, a *Bruker* AXS D8 advance diffractometer in Bragg–Brentano geometry equipped with Cu-Kα radiation with a Sol-X energy dispersive detector; and (b) for high resolution analysis, carried out on selected samples, Synchrotron Radiation X-ray Diffraction (SR-XRD) data were collected at the high-resolution station (beam line BM01B) at the Swiss-Norwegian (SNBL) at ESRF (Grenoble, France). For the latter, the wavelength was 0.5002 (1) Å. For the measurements in both instruments, the powder samples were packed and sealed in boron-glass capillaries with 0.8 mm inner diameter under clean argon atmosphere in a glove box. The capillaries were rotated during the measurements to improve the powder averaging.

A full profile fitting procedure based on the Rietveld method, implemented in the software package GSAS with the user interface EXPGUI [28] was employed to obtain microstructural information such as crystallite size. The crystallite size approach used here was based on the Peter Stephens formalism implemented in the profile function 4 (*TCp-x*) described in the GSAS Manual [29].

The hydrogen desorption behavior from the powder mixtures was investigated by differential scanning calorimetry (DSC) measurements performed in a NETZCH STA 449 F3 using a heating rate of 5 K/min under Argon atmosphere from room temperature (RT) to 500 °C. Samples with mass around 15 mg were used.

The H-absorption/desorption properties were evaluated in a volumetric Pressure-composition-temperature (PCT) apparatus designed and built in-house at Institute for Energy Technology (IFE), using samples with mass of around 100 mg. Absorption and desorption curves were measured at 325 °C and at a pressure of 2 MPa and 0.05 MPa for absorption and desorption, respectively.

Scanning electron microcopy (SEM) images coupled with energy-dispersive X-ray spectroscopy (EDS) data were obtained in a Magellan 400L (MEV-FEG FEI) microscope.

Results and discussion

Microstructural characterization: SEM and XRD analysis

Fig. 1 shows SEM micrographs obtained in back-scattered electrons (BSE) mode for MgH_2 with 2 mol.% FeF_3 and NbF_5, respectively. Fig. 2 presents the EDS spectra in selected regions for the cryomilled mixtures containing these fluorides additives.

The morphology of the cryomilled materials show a heterogeneous particle size distribution of MgH_2 particles (see Fig. 1): (i) a coarser distribution of particles with average size around 10–25 μm; and (ii) a fine distribution of particles with average size around 0.25–3 μm, which are agglomerated over the surface of the bigger particles. This heterogeneous particle size distribution of MgH_2 particles has been previously observed by Chan et al. [26] and by Chaudhary et al. [27] for MgH_2 – based samples after cryomilling. The SEM analysis for the other compositions have indicated the same features shown in Figs. 1 and 2, and for this reason they are not shown here.

Close evaluation of the fine distribution of particles for the mixture containing FeF_3 (Fig. 1(c)), shows very fine particles, which appear brighter in the micrograph. These particles possibly refer to the Fe from the FeF_3 additive (sizes around 0.2–0.4 μm) which is found randomly dispersed over the surface of the large MgH_2 particles, as confirmed by the EDS spectrum shown in Fig. 2(a).

These morphological aspects, such as the high specific surface area combined with the fine dispersion of additive particles over the MgH_2 particles are known to contribute to the enhancement of the H-absorption/desorption kinetic

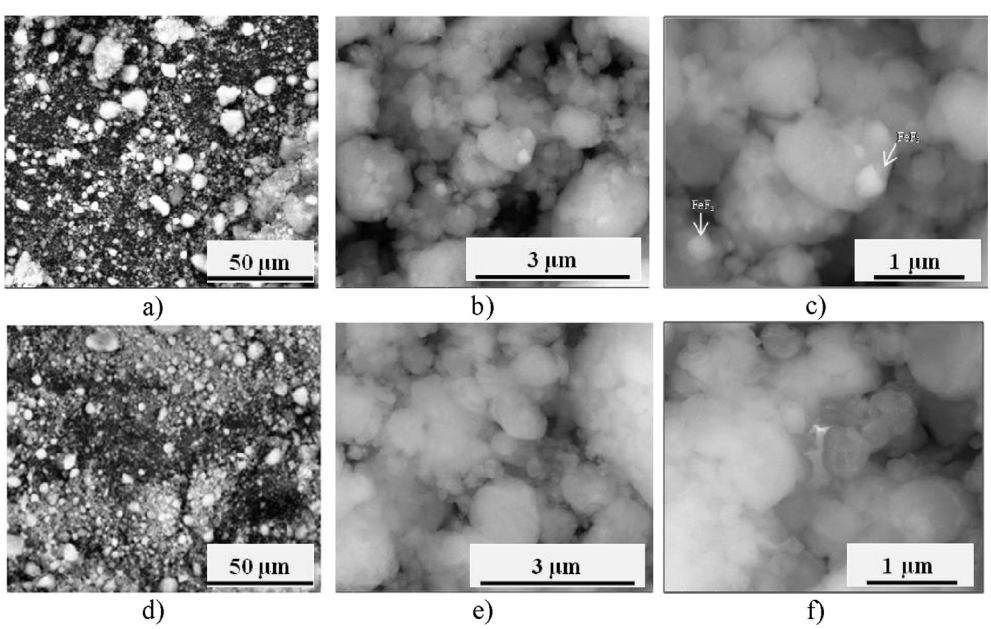

Fig. 1 – SEM images: (a) to (c) for MgH_2 + 2 mol.% FeF_3; (d) to (f) for MgH_2 + 2 mol.% NbF_5.

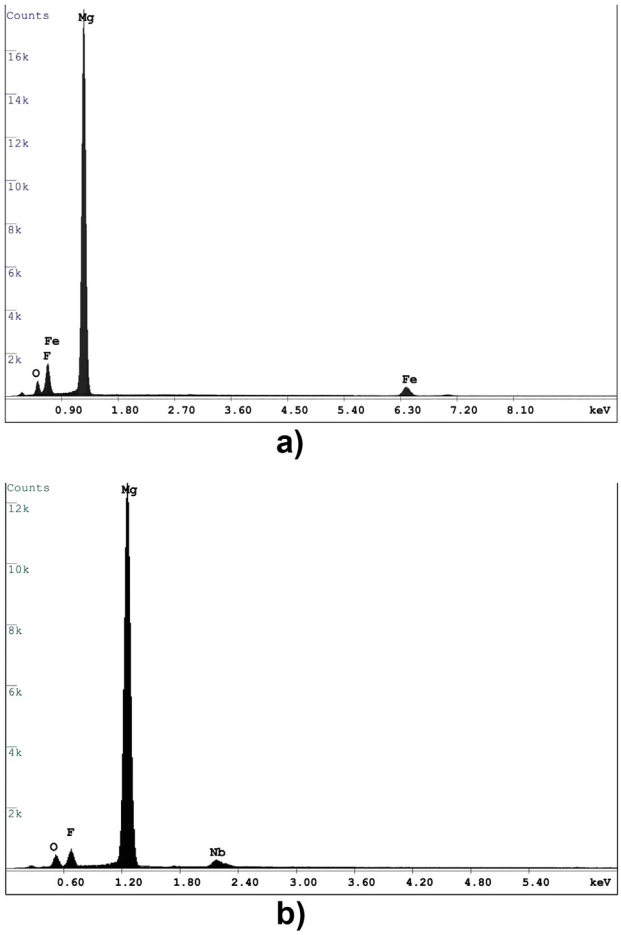

Fig. 2 – EDS spectra; (a) for the mixtures MgH_2 + 2 mol.% FeF_3; (b) for the mixture MgH_2 + 2 mol.% NbF_5.

properties of the MgH_2–FeF_3 mixtures. An analogous positive effect has been previously observed by Hanada et al. [30] during HEBM of MgH_2 containing Ni-nanoparticles as additive.

Although no particles of NbF_5 can be observed in the BSE images of Fig. 1(d)–(f), the EDS spectrum presented in Fig. 2(b) shows that the mixture is entirely composed by Mg, Nb and F (beyond O) and that the additive particles were possibly crushed to the nanometer scale during the milling process [31].

The Cu-Kα XRD pattern of the as-received MgH_2 shows very sharp Bragg peaks corresponding to the β-MgH_2 (tetragonal) and Mg (hexagonal) phases (see Fig. 3(a)). Fig. 3(b) and (c) show the XRD patterns for pure cryomilled MgH_2, MgH_2–Fe and MgH_2–Nb mixtures after the cryomilling process. In these data, the most intense reflections are mainly from the starting materials; β-MgH_2 and the additives Fe, Fe_2O_3, FeF_3, Nb and Nb_2O_5·In addition, weak peaks corresponding to residual Mg are observed. This Mg is likely coming from the starting material as suggested by the XRD pattern of the as-received MgH_2 sample (Fig. 3(a)). Traces of the orthorhombic high-pressure γ-MgH_2 phase are observed in all processed samples. The presence of the metastable γ-MgH_2 phase is well known in pure MgH_2 and MgH_2-based compounds [7,9,14] obtained by HEBM experiments carried out at RT and also more recently, during cryomilling [27].

Some traces of MgO phase were also observed in all mixtures. This is likely due to slight oxygen contamination coming from to the unreacted Mg and found in the starting material. This probably happened due to the not perfect sealing of the capillary.

The XRD patterns of the MgH_2–Nb mixtures (Fig. 3(c)), did not show presence of any intermediate metastable phases like γ/β-NbH_x (x = 0.8–2). It has been observed that during high energy milling of MgH_2 with 3-d transition metals, especially Nb, the formation of Nb hydrides has an effect of mechanical alloying [33].

For the fluoride-containing samples, no MgF_2-like phases, indicating a possible transfer reaction between MgH_2 and F, was observed after milling, and thus in contrast to earlier reports [14,21].

The comparison between the XRD patterns of the cryomilled MgH_2 sample with the samples containing additives reveals a broadening of the β-MgH_2 phase peaks together with a decrease of their intensities. This feature can be clearly seen in Fig. 4(a) and (b) which show details for the strongest Bragg peak of β-MgH_2. The peak broadening is a clear indication that, upon milling the crystallite size is significantly reduced to the nanometer scale. Furthermore strain and structural defects are introduced into MgH_2. This peak broadening is more pronounced for the samples containing fluorides followed by the samples with oxides and transition metals in comparison with the cryomilled MgH_2 sample (Fig. 4(a) and (b)) and also for the as-received MgH_2 sample. The full-width-half-maximum (FWHM) parameters for all samples are shown in Table 2. The FWHM parameter was obtained from the maximum peak of β-MgH_2 phase for $2\theta = 27.4°–28.4°$.

As seen in Table 2, the samples containing oxides and fluorides additives showed larger FHWM parameters, which can be related to a lower crystallite size in comparison with the samples containing metals. The samples containing fluorides showed the biggest broadening among all samples.

These differences could be related to the different milling properties of metals, oxides and fluorides, which may lead to a finer microstructure in the case of oxides/fluorides when compared to pure metals [21–23]. Our study clearly demonstrates that the addition of oxides/fluorides (even in small amounts) can also act as a lubricant, dispersing and/or cracking agent during milling, and thus helping to further reduce MgH_2 particle size during milling. Therefore the lubricant, dispersive and hardness properties of oxides/fluorides can facilitate a further reduction of MgH_2 particle size during milling. Kwon et al. [23] have shown similar results for magnesium milled with Fe_2O_3. On the other hand, metallic additives may be more difficult to disperse due to their ductile behavior limiting the fracture of fine particles [7,33].

In order to investigate the effect of the fluorides acting as crystallite size reducer agent, the fluoride containing samples were selected to further measurement by high-resolution SR-XRD and compared with the pure cryomilled MgH_2. These SR-XRD patterns were then analyzed by the Rietveld method for crystallite size determinations. The strain contribution is too

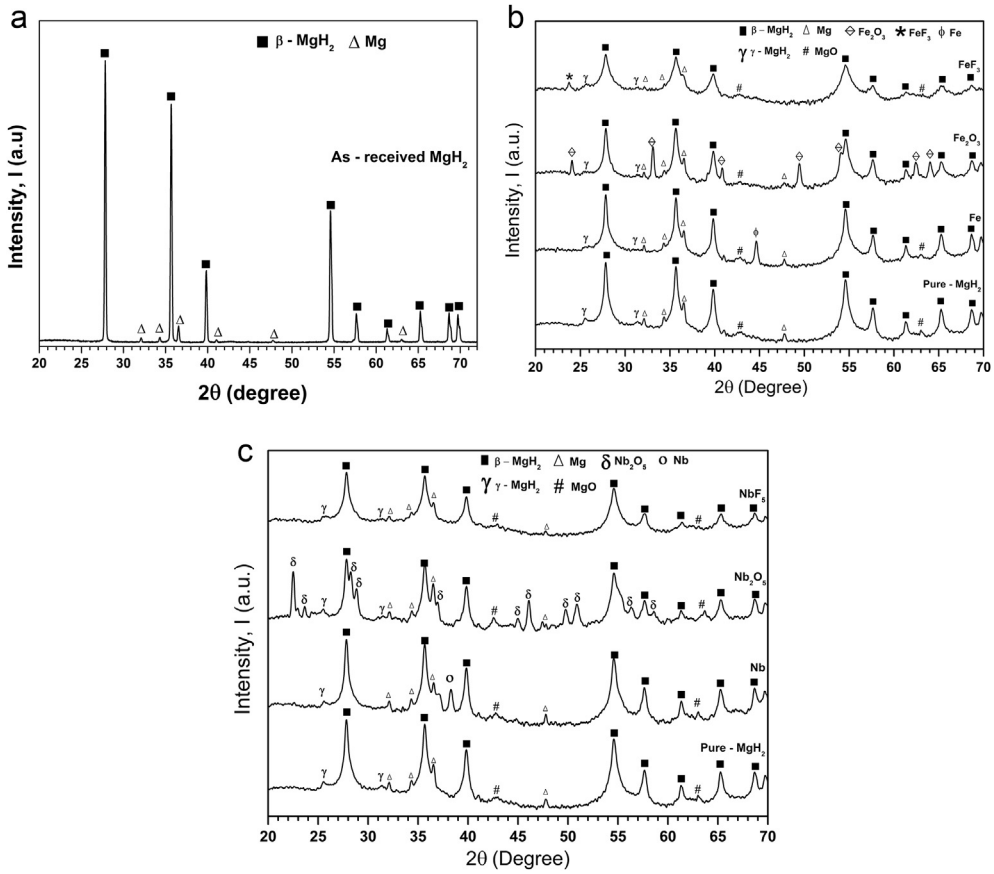

Fig. 3 – XRD patterns (Cu-Kα) of the samples: (a) as-received MgH_2 before the cryomilling process, (b) MgH_2–Fe mixtures (2 mol.% Fe, $Fe_2O_3$and FeF_3) and (c) MgH_2–Nb mixtures (2 mol.% Nb, Nb_2O_5 and NbF_5) after the cryomilling process. The XRD pattern of the cryomilled MgH_2 without any additive (pure) is included in (b) and (c).

small compared to the particle size effect and for this reason the strain was not included in the model.

Fig. 5 shows the SR-XRD patterns for the pure cryomilled MgH_2 and the mixtures of MgH_2+ 2mol.% FeF_3 and MgH_2 + 2mol.% NbF_5, respectively. The SR-XRD patterns basically show all phases already observed in the lab-source XRD patterns (Fig. 3(b) and (c)) except for the fact that the MgO phase was not identified in the samples containing fluorides. This probably occurs due to the better sealing of capillary used for measurement.

The crystallite sizes of β-MgH_2 phase from the SR-XRD patterns as determined by Rietveld analysis are given in Table 3. The mixtures containing fluorides display smaller crystallite sizes compared to the pure sample and the mixture containing FeF_3 has the smallest crystallite size among the samples (as already indicated by FWHM parameter in Table 2). The crystallite size obtained for the cryomilled MgH_2 sample is around 29 nm while for the mixtures containing NbF_5 and FeF_3 the crystallites sizes are around 13 nm and 11 nm, respectively. The crystallite sizes of the fluorides samples are almost three times smaller than for pure cryomilled sample, and thus showing the strong mechanical effect when fluorides are milled together with MgH_2 resulting in a much finer microstructure. This tendency is in agreement with ref. [31].

The milling conditions have a big influence in the final material, especially with respect to crystallite size [34]. The values for the average crystallite size of β-MgH_2 phase of the cryomilled samples are slightly bigger than published earlier [34,35] with Mg/MgH_2 mixtures ball milled under argon and hydrogen atmosphere. However, considering the milling process under Ar performed at short milling times (~15 min) in mixture of MgH_2 + NbF_5 [36] our values for crystallite sizes are significantly smaller (20 nm vs. 13 nm).

H-absorption/desorption characterization: DSC and kinetics analysis

Fig. 6 shows the DSC curves for all the samples after the cryomilling. The DSC curve of the as-received MgH_2 powder is also included for comparison. The peak desorption temperature is indicated in each curve. The DSC curves for all processed samples show a significant decrease in the peak desorption temperature compared with the as-received powder. The initial as-received MgH_2 exhibits a single endothermic peak at 418 °C related to the release of hydrogen.

The effect of milling at low temperature has produced a significant result in reduce the desorption temperature of MgH_2 (around 60 °C of reduction), as could be seen by comparing the as-received sample with the cryomilled MgH_2. Now, considering the mixtures samples, the catalytic effect of the additives is clear; all the mixtures containing additives presented a decrease in the peak desorption temperature as

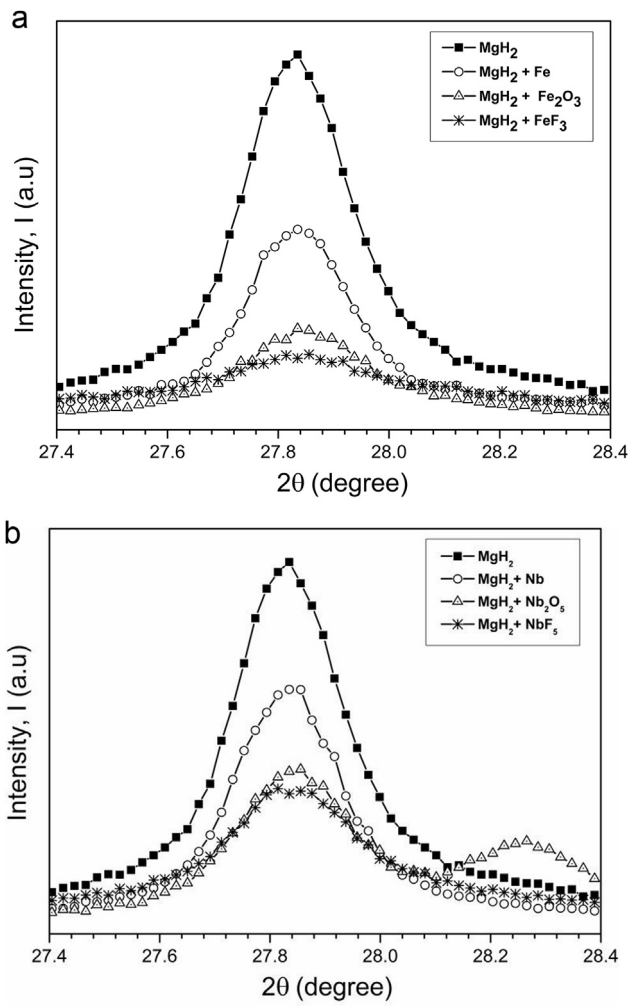

Fig. 4 – XRD patterns (Cu-Kα) of MgH_2–Fe and MgH_2–Nb mixtures at range of 2θ = 27.4° to 28.4°. The cryomilled MgH_2 sample without any additive is also included.

compared with the pure cryomilled MgH_2 sample. The efficiency of this beneficial effect varies according to the type/nature (metal, fluoride or oxide) of the additive. The following sequence is observed from the highest to lowest desorption temperature: transition metals (Fe and Nb), oxides (Fe_2O_3 and Nb_2O_5) and fluorides (FeF_3 and NbF_5). The peak desorption temperatures observed for the transition metals (Fe and Nb)

Table 2 – Values of FWHM for all processed samples. The as-received MgH_2 is also included.

MgH_2 samples	FWHM (full width at half maximum)
As-received	0.1413
Pure[a]	0.2425
Fe	0.2316
Fe_2O_3	0.3412
FeF_3	0.7000
Nb	0.2240
Nb_2O_5	0.2380
NbF_5	0.3680

[a] MgH_2 sample cryomilled without any additive.

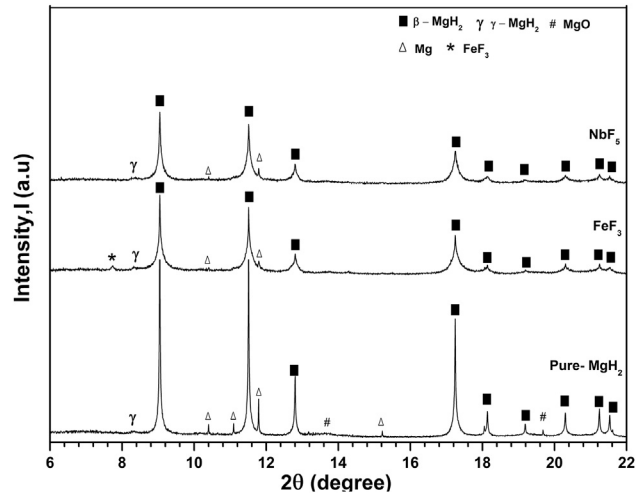

Fig. 5 – SR-XRD patterns (λ = 0.5002 Å) of pure cryomilled MgH_2 and for the mixtures of MgH_2 + 2 mol.% of FeF_3 and NbF_5, respectively.

are close to the pure cryomilled MgH_2 while for the oxides it is a significant decrease in the desorption temperature compared to pure cryomilled MgH_2, 21 and 61 °C lower for Fe_2O_3 and Nb_2O_5 additions, respectively. The biggest reduction in the desorption temperature reduction is for the mixtures containing fluorides; with around 81 °C in the case of FeF_3 and 83 °C in the case of NbF_5 addition. These mixtures start to decompose around 198 °C (onset temperature) and the desorption is completed at ~315 °C.

A comparison among each of the additives groups shows that the effects on desorption temperature of metals (Nb and Fe) and fluorides (FeF_3 and NbF_5) are very similar. For the oxides it is a significant difference. The Nb_2O_5 is superior to Fe_2O_3 as also reported in Ref. [16].

The catalytic effect of oxides (e.g. Fe_2O_3 and Nb_2O_5) and fluorides (e.g. FeF_3 and NbF_5) has been extensively studied by HEBM performed at RT [16,31,37]. Nb_2O_5 has been reported to be much more effective than the corresponding fluoride NbF_5 in decreasing the desorption temperature of MgH_2 [10]. In our results we have observed the opposite where the effect of NbF_5 is slightly superior to their oxide. Furthermore, when cold rolling was used to process similar MgH_2-based mixtures used in the present work [12], the reduction observed for the desorption temperature was smaller than our results shown in Fig. 6. This could be associated to the degree of dispersion of additives into MgH_2 matrix, as well as to the refinement attained by the different processing routes.

In addition, the DSC curves for the samples containing Nb, Nb_2O_5 and Fe_2O_3 exhibited a shoulder at higher temperatures

Table 3 – Crystallite sizes for β-MgH_2 phase as determined by Rietveld analysis for: pure cryomilled – MgH_2, MgH_2 + 2 mol.% FeF_3 and MgH_2 + 2 mol.% NbF_5.

	Crystallite size (nm)
MgH_2 – pure	29(2)
MgH_2 + FeF_3	11(3)
MgH_2 + NbF_5	13(2)

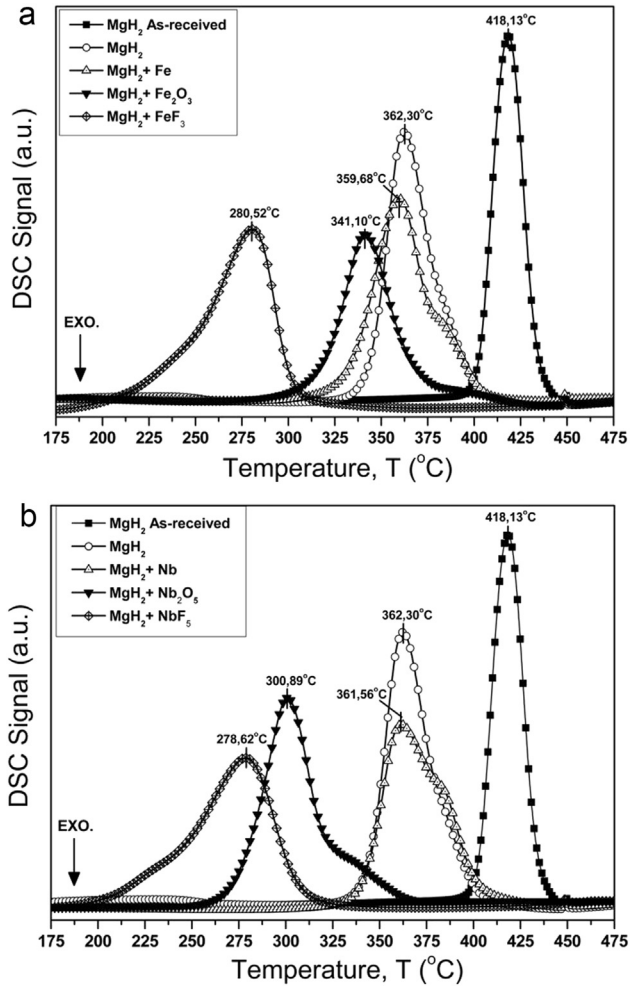

Fig. 6 – DSC curves for the mixtures: (a) MgH_2–Fe mixtures (2 mol.% Fe, Fe_2O_3 or FeF_3) and (b) MgH_2–Nb mixtures (2 mol.% Nb, Nb_2O_5 or NbF_5) after the cryomilling process. DSC curves of pure cryomilled MgH_2 and as-received MgH_2 are also included. The peak desorption temperatures (°C) are indicated for each sample.

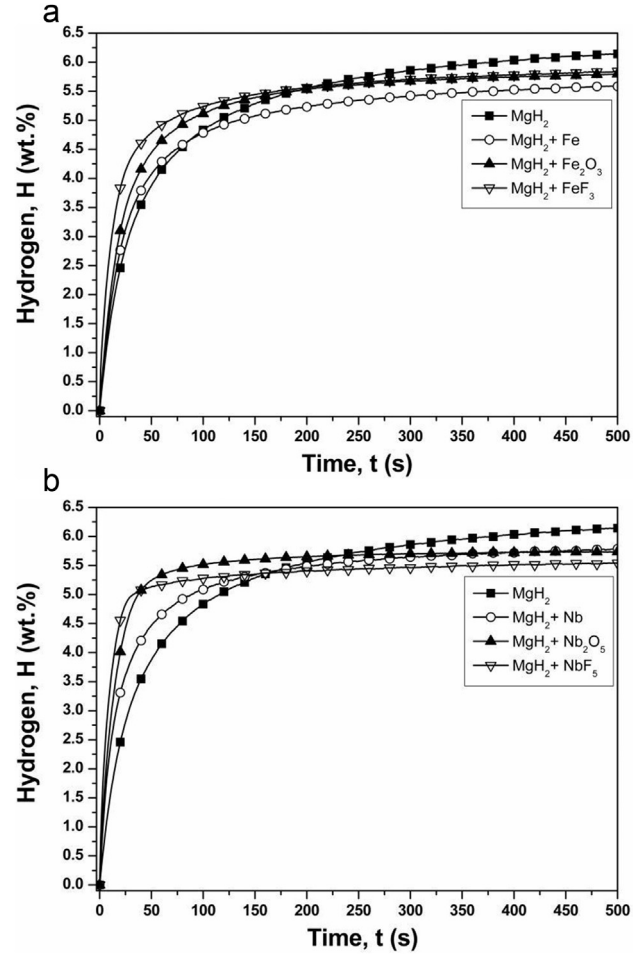

Fig. 7 – Hydrogen absorption kinetics at 325 °C under 2 MPa of H_2 pressure of the samples: (a) MgH_2–Fe and (b) MgH_2–Nb mixtures after cryomilling process. The pure cryomilled MgH_2 is shown for comparison.

after the first endothermic peak. This shoulder could be associated mainly with the inhomogeneous distribution of the particle size [34] observed by SEM analysis for MgH_2 and the additive resulting from the cryomilling process. This feature likely leads to multiple steps in dehydrogenation kinetics of MgH_2.

Another possible explanation for this double DSC peak is the coexistence of γ and β phases and their different decomposition steps during hydrogen desorption [38]. In this case, the first peak corresponds to the β (partial) and γ desorption and the second peak to the β phase.

Fig. 7 displays the absorption kinetics curves measured at 325 °C under 2 MPa of H_2 for the MgH_2–Fe and MgH_2–Nb mixtures, respectively. The data for pure cryomilled MgH_2 is included for comparison. It can clearly be seen that all mixtures have faster absorption kinetics than the pure sample. In addition, the Nb-containing additives presented a superior effect in comparison with the Fe-containing group. The mixtures containing the fluorides NbF_5 and FeF_3 showed the fastest absorption kinetics, followed in order by the mixtures with the oxides Nb_2O_5, Fe_2O_3 and the transition metals Nb, Fe. The complete hydrogen absorption of around 5.5 wt.% of H_2, is reached in 70 s for the sample containing NbF_5 while 150 s are needed for the mixture with FeF_3. These results are in agreement with the DSC curves analysis.

The desorption kinetics curves measured at 325 °C under 0.04 MPa of H_2 for the MgH_2–Fe and MgH_2–Nb mixtures are shown in Fig. 8. Overall, the desorption kinetics from all mixtures are significantly improved compared to pure cryomilled MgH_2.

The best desorption kinetics are observed in the mixtures containing the fluorides NbF_5 and FeF_3, followed in order by the oxides Nb_2O_5, Fe_2O_3 and the transition metal Nb. No incubation time was observed for the mixtures containing NbF_5, FeF_3 and Nb_2O_5 and the fully desorption is reached after 1700 s. Furthermore, the effect of additions of Nb_2O_5, NbF_5 and FeF_3 seems to be very similar with respect to the desorption kinetics.

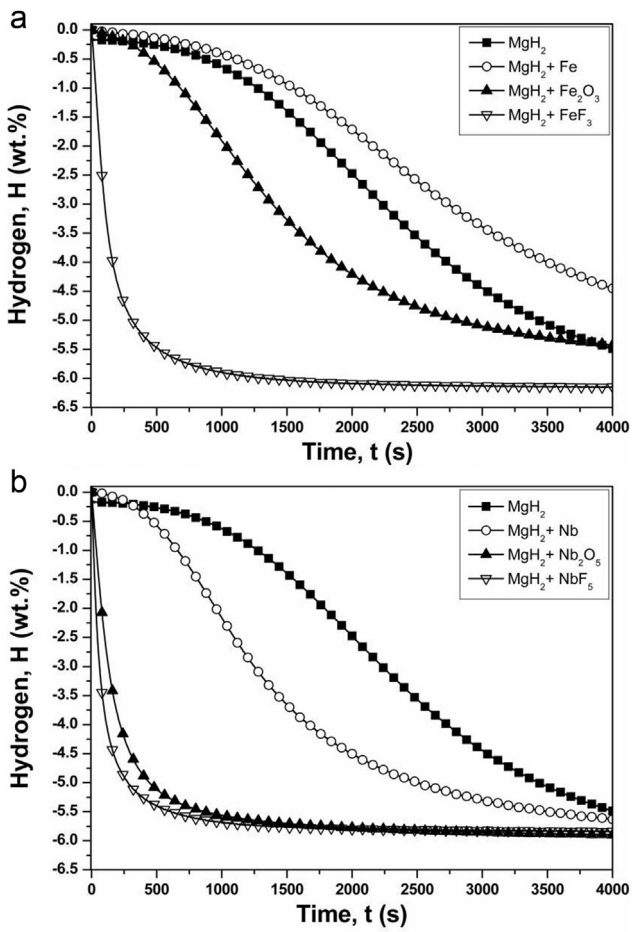

Fig. 8 – Hydrogen desorption kinetics at 325 °C under 0.04 MPa of H_2 pressure of the mixtures: (a) MgH_2–Fe; and (b) MgH_2–Nb mixtures after cryomilling process. The pure cryomilled MgH_2 is plotted for comparison.

On the other hand, the desorption kinetics curve for the mixture with Fe is slower than for the pure cryomilled sample. This can be related to the ductile behavior of Fe additive during milling with MgH_2. The addition of Fe into Mg promotes a cold welding mechanism inducing agglomeration of the metal particles during milling [14]. Beyond in this case the temperature during milling is not sufficiently low to promote embrittlement of the metals.

In our particular case, we observe that the effect of Fe is more effective for absorption than for desorption.

Our 3 h cryomilled sample with FeF_3 additive shows a similar desorption kinetics as for 48 h of milling under argon of MgH_2 with 5 mol.% FeF_3 [37]. Furthermore, the superior effect on desorption kinetics behavior of MgH_2 with 0.2 mol.% of Nb_2O_5 addition was observed in Ref. [10], which reports a full desorption after 140 s but after 120 h of milling under argon atmosphere.

As for NbF_5 addition to MgH_2, the studies [31,36] have already showed that the addition of NbF_5 significantly improves the sorption kinetics of MgH_2 (consistent to our results). Thus it would be an ideal additive to confirm whether or not nanosized and surface oxide-free Nb hydrides particles will act as an effective catalyst and if they are formed in situ during ball milling and/or during the hydrogenation processes.

More recently, the superior effect of Nb_2O_5 and FeF_3 in comparison with Fe, Nb and Fe_2O_3 additions (2 mol.%) on desorption kinetics behavior in cold rolled MgH_2 samples has been observed [13].

The remarkable improvements observed in the hydrogen sorption properties with both low desorption temperature and fast absorption/desorption kinetics can be understood by the presence of active nucleation sites created by additive inclusions creating abundant interfaces with MgH_2. This is beneficial for the catalysis and hydrogen desorption property of MgH_2. Furthermore, the crystallite size reduction to the nanometer scale attained during cryomilling was well evidenced when fluorides were used as additives. The fluorides promote crystallite size reduction of β-MgH_2 as confirmed by SR-XRD analysis followed by Rietveld refinement. Another interesting feature helping to explain the excellent properties observed for the mixtures containing fluorides is the fine additive dispersion (especially for FeF_3 particles) over the MgH_2 matrix, as indicated by the SEM analysis. These aspects are in agreement with [39,40] for milling processes conducted at RT.

Conclusions

The following conclusions could be drawn from this investigation:

- SEM and XRD analysis have shown a very heterogeneous particle size distribution accompanied by smaller crystallite size for the samples containing fluorides (in comparison with the pure cryomilled sample). Furthermore, the γ-MgH_2 phase was barely observed in all samples after milling at low temperature.
- Crystallite size determination from the SR-XRD data has revealed that there is a strong mechanical effect when fluorides are milled together with MgH_2 resulting in a much finer microstructure. We conclude that the fluorides act as a lubricant, dispersing and/or cracking agent during milling, helping to further reduce the MgH_2 crystallite size.
- The strong catalytic effect of the additives is shown from the reduction of the desorption temperature in the DSC data. The best results were observed for fluorides, oxides and transition metals, respectively. Among the oxides, Nb_2O_5 is more efficient than Fe_2O_3.
- The absorption and desorption kinetics curves for samples containing additives reveal a superior performance in comparison with the pure cryomilled sample. The best results for desorption kinetics were observed for the fluorides: NbF_5 and FeF_3 (equivalent effect and consistent to the DSC analysis) followed by the oxides: Nb_2O5, Fe2O3 and Nb. The addition of Fe was not efficient in comparison with the pure sample.
- Based on the DSC analysis, kinetics and microstructural aspects present here the importance of finer microstructure achieved via the mechanical interactions between additives with MgH_2 during milling is reinforced.
- Finally, we could observe that similar positive effects (decreasing desorption temperature and improving

sorption kinetics) of the additives (studied here) in MgH_2 during milling at low temperature are also observed when milling is performed at room temperature.

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